# ADF, Databricks & DBT Hands-on instructions

SOLITA

# Introduction: What you'll build

You will create a complete cloud resources, data warehouse using Azure, Databricks and Unity Catalog:

- ✓ Azure Resource group
- ✓ Azure Resource Group with ADLS Gen2, Key Vault, and Azure Data Factory
- ✓ ADF pipeline to ingest data from Statistics Finland (StatFin) API
- ✓ Raw CSV files stored in ADLS organized by dataset
- ✓ Databricks workspace connected to your storage via external locations
- ✓ Bronze layer tables in Unity Catalog via Python notebooks
- ✓ Silver, Gold, and Publishing layer transformations via dbt
- ✓ Genie AI-powered natural language queries (extra credit)

**Prerequisites:**

- Azure account with sandbox subscription access
- Access to portal.azure.com
- Databricks workspace and Unity Catalog already available (shared sandbox)
- Facilitator available to assign external locations in Databricks

# Step 1: Create Azure Resource Group and Resources

## Instructions

- Navigate to portal.azure.com and sign in with your sandbox account
- Create a Resource Group
- Inside this Resource Group, create the following resources (all in North Europe):
    - Storage Account (ADLS Gen2)
    - Key Vault
    - Azure Data Factory
- Once all resources are created, verify they all appear in your Resource Group

Note: Use the same region (North Europe) for all resources to avoid data transfer issues.

# Step 2: Set Up ADLS Folder Structure

## Instructions

- Navigate to your Storage Account in the Azure portal
- Create a container, followed by folder structure for the datasets.

  Verify your folder structure looks like this for example :

```
raw-data/
        ├── bankruptcies/
        ├── population/
        └── enterprise-establishments/
```

Note: These folder paths will be used as sink destinations in your ADF pipeline.

# Step 3: Add Secrets to Key Vault

## Instructions

- Navigate to your Key Vault in the Azure portal
- Create the following 3 secrets one by one:
  - Secret 1 - For storage account name
  - Secret 2 - For storage account key
  - Secret 3 - For statfin url
- After creating all 3 secrets, verify they appear in the Secrets

Note: ADF will reference these secrets via Key Vault linked service — this avoids hardcoding credentials in your pipeline.

# Step 4: Create Linked Services in ADF

## Instructions

- Open your Azure Data Factory → Click: Launch Studio

- Create linked services:
- Linked Service 1: For Azure Blob Storage / ADLS Gen2 and test the connections

- Linked Service 2: For HTTP (StatFin API) and test the connections

- Verify both linked services appear in the Linked Services list

Note: Always test each connection before moving to dataset creation.

# Step 5: Create Datasets in ADF

## Instructions

- Go to: Author → Datasets → + New Dataset
- Dataset 1: Create dataset for Source — StatFin HTTP and test connections
- Dataset 2: Create dataset for Sink — ADLS Blob Storage and test the connections
- Note: Both datasets will use parameters passed from the pipeline — do not hardcode file paths here.

# Step 6: Create the ADF Pipeline

## Instructions

- Go to: Author → Pipelines → + New Pipeline
    - Name: PL_StatFin_IngestionDataset

- Create Pipeline Parameters
    - parameter: Name: loopParameters Type: Array
    - This array will hold the list of source/sink config objects from your JSON config file

- Add a ForEach Activity
    - Items: @pipeline().parameters.loopParameters

- Inside the ForEach activity (double-click to open):
    - Add a Copy Data activity
    - Name it: Copy_StatFin_To_ADLS

- Configure Copy Activity with required source and sink tab details

- Configure Copy Activity — Sink tab:

# Step 7: Validate, Publish and Run the Pipeline

## Instructions

- Validate the Pipeline
  - Fix any errors shown in the validation output panel
  - Common issues: missing parameters, incorrect dataset references
  - Re-validate until you see: "Your pipeline has been validated. No errors were found."

- Publish the Pipeline
  - All pending changes to linked services, datasets, and pipeline will be saved

- Trigger the Pipeline (Manual Run)

- Monitor the Pipeline Run
  - Wait for the status to show: Succeeded

- Verify Output Files in ADLS

Note: All 3 files must be present before proceeding to Databricks.

# Step 8: Launch Databricks and Connect Storage

## Instructions

- Open the Databricks workspace
  - Unity Catalog is already enabled and available (catalog: data_academy_fi_06)

- Get External Location assigned by Facilitator
  - Contact your facilitator and provide:
  - Your storage account name
  - Your container name: raw-data
  - The facilitator will create an External Location in Unity Catalog pointing to your ADLS storage account
  - This allows Databricks to access your files via the abfss:// protocol

- Grant permissions to your storage (if not already done by facilitator)
  - Role: Storage Blob Data Contributor
  - Assign to : the Databricks managed identity or your user account

- Verify the External Location in Databricks

- Note: Do not proceed to notebook creation until the external location test passes.

# Step 9: Create Schemas in Unity Catalog

## Instructions

- In Databricks workspace go to: Workspace → Create → Notebook
- Create a schema required using SQL.
- Verify all 4 schemas appear in the output
- Also verify in the Catalog Explorer:
    - Go to: Catalog → data_academy_fi_06
    - You should see your 4 schemas listed
- Naming convention reminder:
    - Schema: <yourname>_statfin_bronze / <yourname>_statfin_silver / <yourname>_statfin_gold etc
    - Tables: bronze_statfin_<tablename> , silver_statfin_<tablename> etc

Note: Replace <yourname> with your actual first name, lowercase, no spaces.

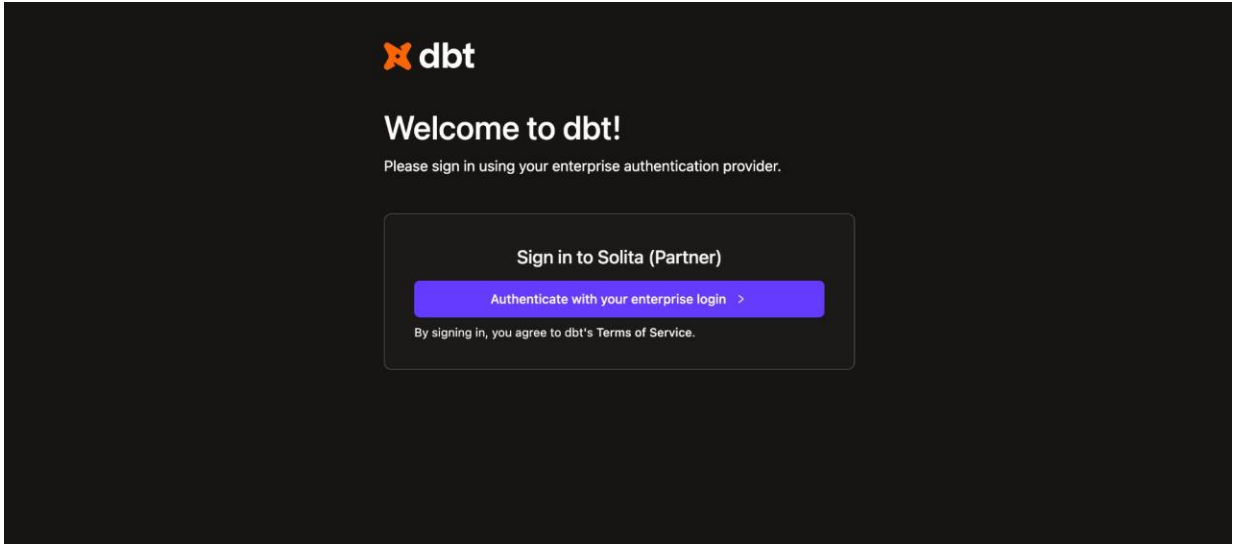
# Step 10: Create Bronze tables

## Instructions

- In Databricks workspace go to: Workspace → Create → Notebook
- Create a bronze tables- data ingestion notebook in required language(python, pyspark etc).
- Create all the bronze tables along with some metadata.
- Verify all bronze table appear under the Bronze schema in databricks.

Note: Replace <yourname> with your actual first name, lowercase, no spaces.

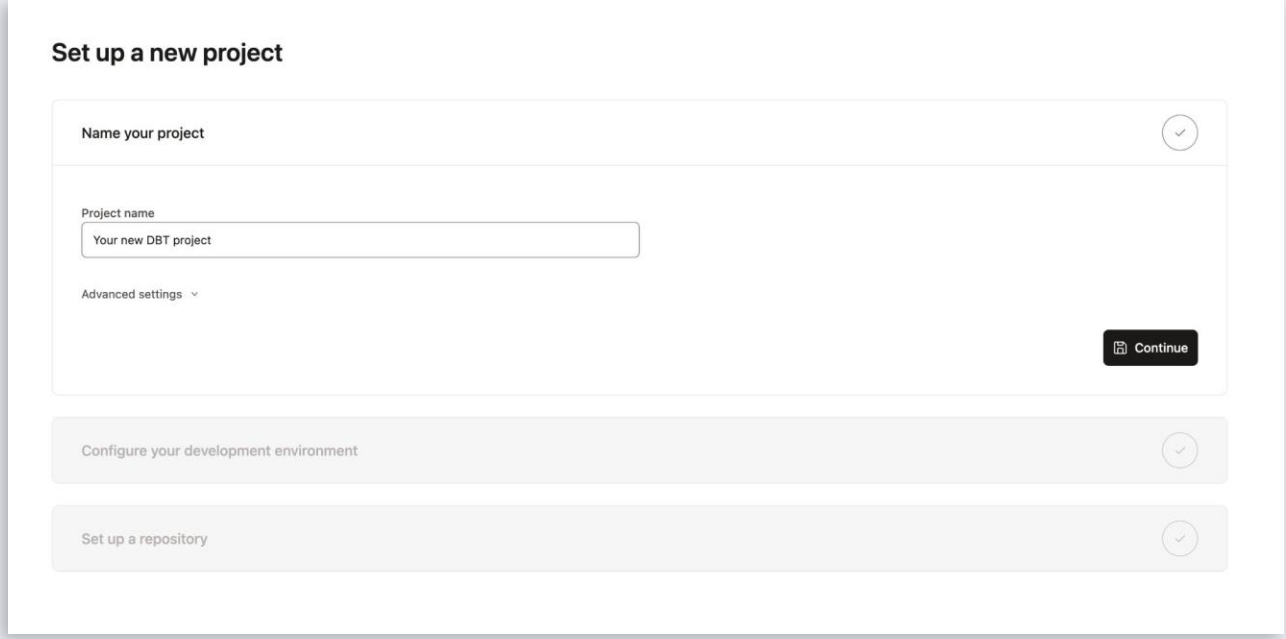
# Step 1: Sign In to dbt Cloud




## Instructions

- Navigate to Solita's dbt Cloud instance
- Click "Authenticate with your enterprise login"
- Use your Solita credentials to sign in via SSO
- You will be redirected to the project setup wizard

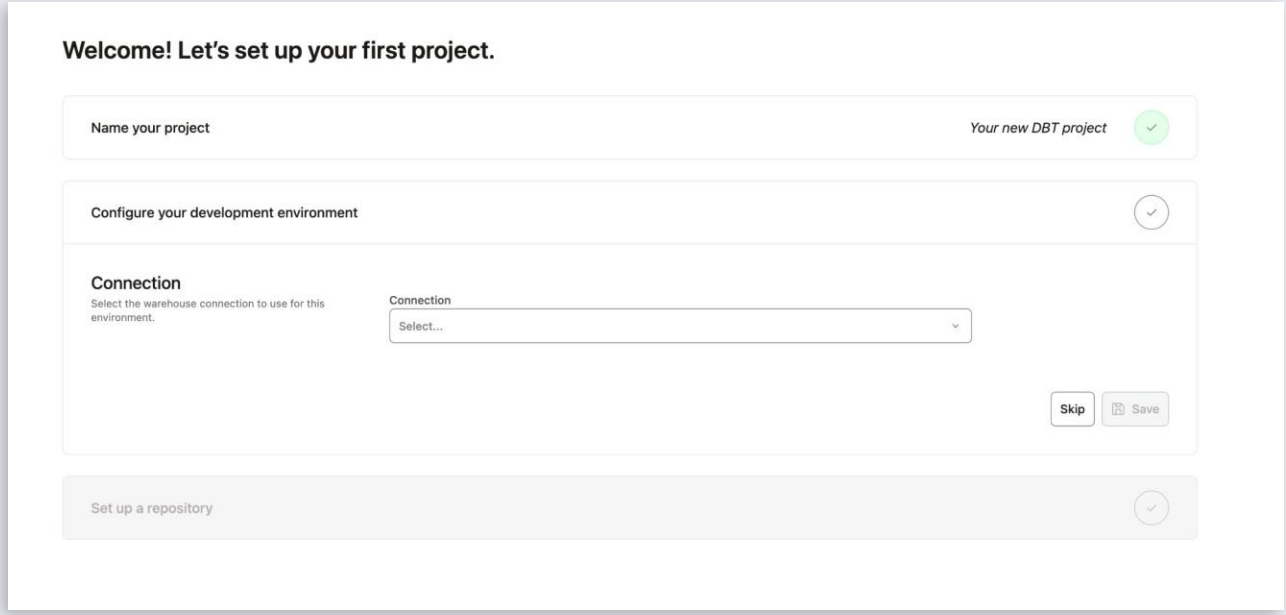
# Step 2: Name Your Project




## Instructions

- Enter a descriptive project name (your name + statfin + databricks)
- Leave Advanced Settings as default
- Click "Continue" to proceed

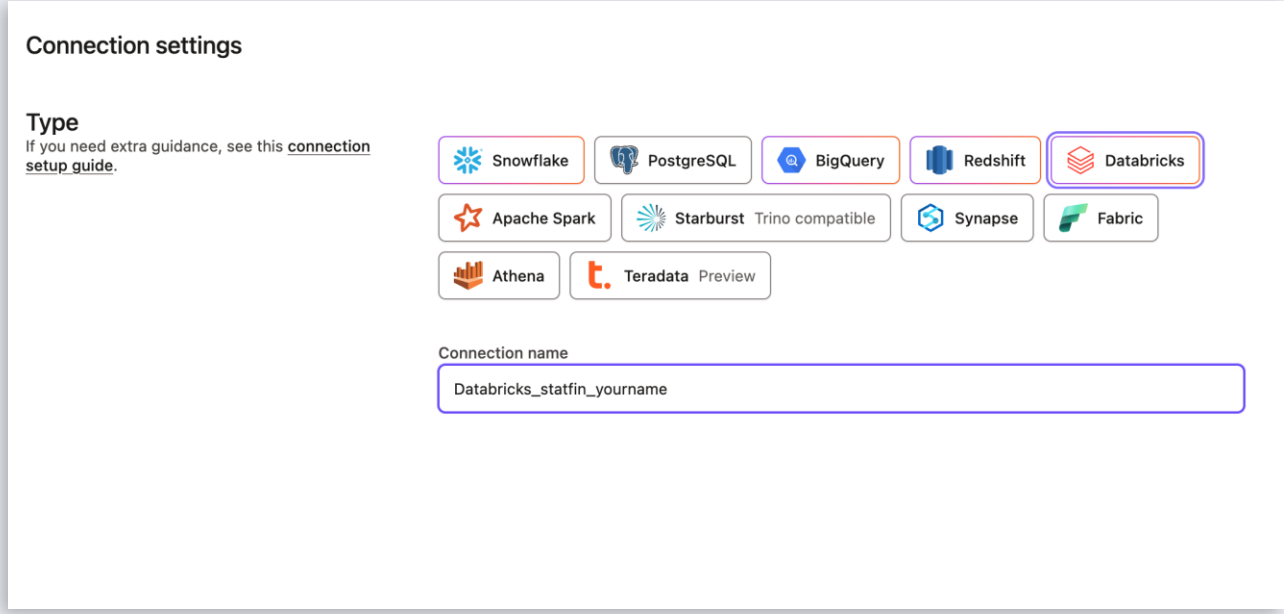
# Step 3: Configure Environment




## Instructions

- Select the warehouse connection from the dropdown
- Select Add New Connection

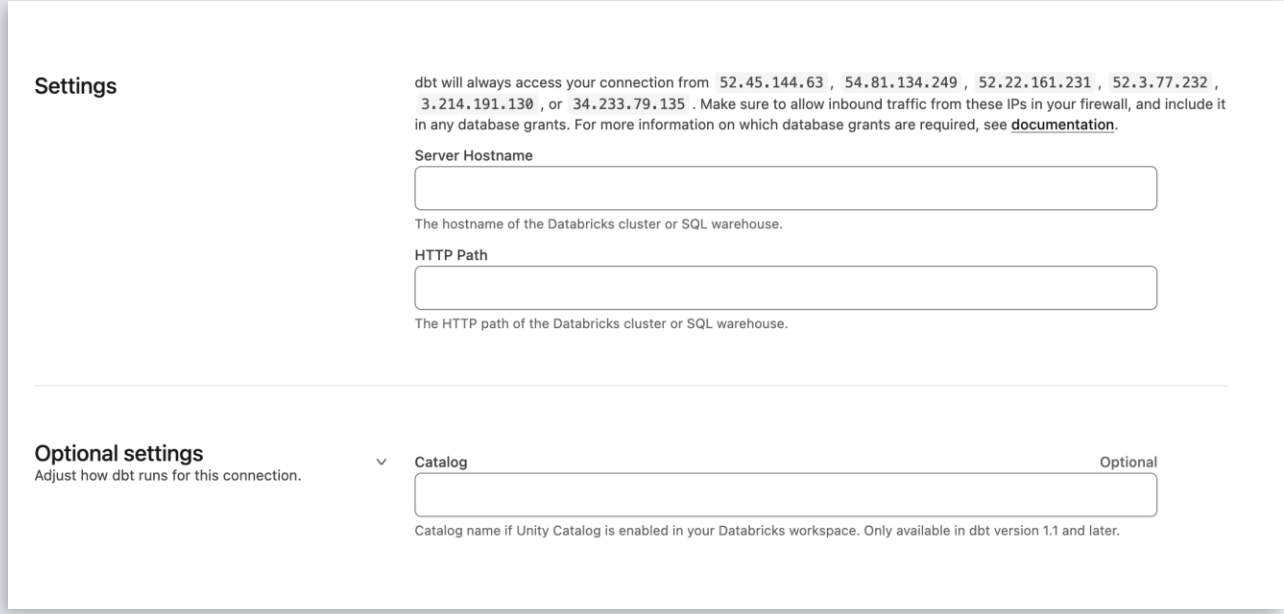
# Step 4a: Connection part 1




## Instructions

- Choose Databricks
- Name your new connection as databricks_statfin_yourname

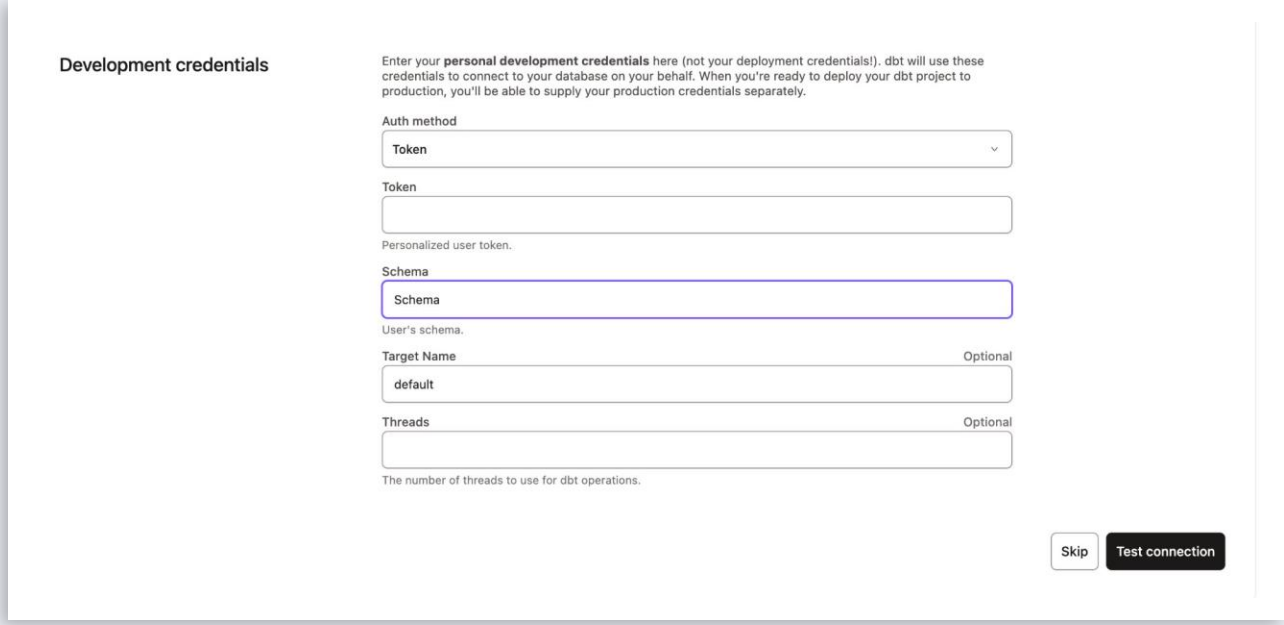
# Step 4b: Connection part 2




## Instructions

- From settings insert your Servers Hostname and HTTP Path. You'll get those from Databricks
- Optional settings Catalog should point to your bronze layer in Databricks

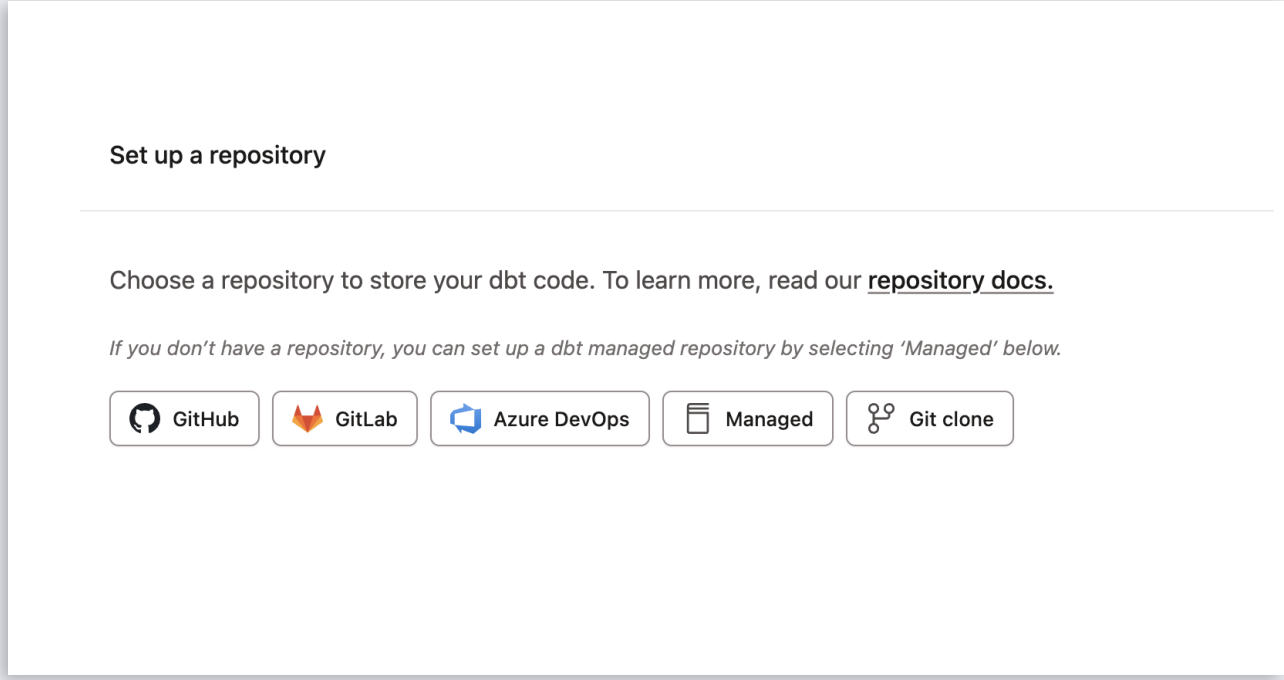
# Step 4: Development Credentials (Databricks)




## Instructions

- Auth method: Token
- Enter your personal access token. You can create that in Databricks
- Set Schema to your personal schema name
- Click "Test connection" to verify

# Step 5: Set Up Repository




## Instructions

- Choose your repository provider
- For CaseWork: select "Managed" for a dbt-managed repo
- Complete setup and start developing!

# **Step 6:** DBT Project Quick setup

- Configure *dbt_project.yml*
  - Set project name and schema names for each layer
  - Define materialization (table / view)
- Create folder structure
  - *models/silver/* and *models/gold/* (or your chosen layers)
- Define sources in *models/silver/sources.yml*
  - Point to your source schema in Databricks Unity Catalog
  - List your source tables (bankruptcies, enterprises, population etc)
- Create silver and gold models
  - Use `{{ source() }}` and `{{ ref() }}` functions to reference data
- Build dbt models and validate
  - dbt run → build models
  - dbt test → run tests
  - dbt build → both in one step

# Step11: Explore Data with Databricks Genie

## Instructions

- What is Genie?
  Genie is Databricks' AI assistant that lets you query your data using plain English — no SQL required. It reads your Gold layer tables and generates queries, charts and summaries from natural language questions.
- Setup:
  - Go to: Workspace → Genie → New Genie Space
  - Add your Gold layer tables: fact_bankruptcies, fact_population, dim_municipality, dim_industry, dim_year
  - Give your Genie Space a name (e.g. StatFin Analytics)
- Sample questions to try:
  - "Which municipality had the most bankruptcies in 2023?"
  - "Show bankruptcy trend by industry from 2020 to 2024"
  - "Compare new enterprise establishments vs bankruptcies by region"
  - "Show top 10 municipalities by total deaths in 2022"
- Creating a Dashboard:
  - Once Genie returns a result you like, click: Pin to Dashboard
  - Give your dashboard a name (e.g. Finland Economic Overview 2020–2024)
  - Add multiple pinned charts to build a full dashboard
  - Go to: Dashboards to view, edit and share your published dashboard
- Dashboard ideas based on your data:
  - Finland Economic Health — bankruptcies vs new enterprises per year
  - Regional Mortality Tracker — deaths and population change per municipality
  - Municipality Growth Scorecard — population vs business activity 2020–2024

- Create your own reports and pin them to dashboard based on your understanding on data.

# Resources

## Documentation

- Unity Catalog:
  https://learn.microsoft.com/en-us/azure/databricks/

- Community forums:
  https://community.databricks.com

- Solita Databricks Sandbox:
  https://adb-7405608897358843.3.azuredatabricks.net/?o=7405608897358843

- DBT portal:

# You're Ready! 🚀

ADF, Databricks & DBT Setup Complete

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Questions? Check the docs or speak to your facilitator.